

HR-FONT: SEPARATING BANK-SUPPORTED GLYPH VARIATION FROM REFERENCE-OBSERVED STYLE FOR CROSS-SCRIPT FONT GENERATION

Anonymous authors

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ABSTRACT

We introduce HR-Font, a conditional diffusion framework for few-shot cross-script font generation. It separates two information sources that prior systems entangle. The task is underdetermined because Chinese references reveal a typeface’s appearance but do not directly reveal how its Latin, Japanese, or phonetic glyphs should be designed. Our *Set-Delta Variation Prior* retains same-character donor glyphs as normalized geometric residuals from a fixed neutral anchor until location-dependent aggregation, while *Graphics-Informed Local Reference Attention* uses cross-script geometric primitives as keys and learned local reference features as values. Together, these modules make ambiguous cross-script completion a controllable design process: the bank supplies a structured variation space, and the references select the appearance that best represents the target family. On [TBD: dataset and evaluation protocol], HR-Font improves [TBD: headline family/style result] while preserving glyph identity at [TBD: identity result]; controlled studies show [TBD: Set-Delta and Graphics-Ref findings].

1 INTRODUCTION

Designing a font across writing systems is not ordinary image translation. Given a small set of Chinese glyphs from an unseen typeface, a model may observe weight, slant, contrast, terminal shape, and texture, yet it does not observe the target Latin letter or Japanese kana itself. Multiple target glyphs can therefore be compatible with the same references. A generator can exploit this ambiguity by combining a broad vocabulary of proven glyph structures with the target family’s visible design language.

Existing few-shot font generators typically combine a neutral content glyph with a global or local style representation (Yang et al., 2023; Tang et al., 2022; Pan et al., 2023). Content-fusion methods additionally blend corresponding characters from basis fonts (Wang et al., 2023). Two failure modes remain exposed in the cross-script setting. First, compressing retrieved same-character examples into one mean feature can cancel distinct structural modes and can transmit donor-specific appearance as if it were target-font evidence. Second, globally pooling a few reference glyphs removes localized cues such as terminals, ligatures, stroke contrast, and dry-brush effects before the generator can use them.

HR-Font addresses the two failure modes with an explicit evidence decomposition (Figure 1). The bank answers *what structural variations are plausible for the requested character*; the observed references answer *which visible appearance is compatible with the target family*. The denoiser combines both sources while the neutral content glyph remains responsible for character identity.

The design principle is simple: the font bank expands structural choice, and target references personalize appearance.

- We establish an evidence architecture for cross-script font completion: *bank-supported structural variation* defines the design space, while *reference-observed appearance* personalizes the result.

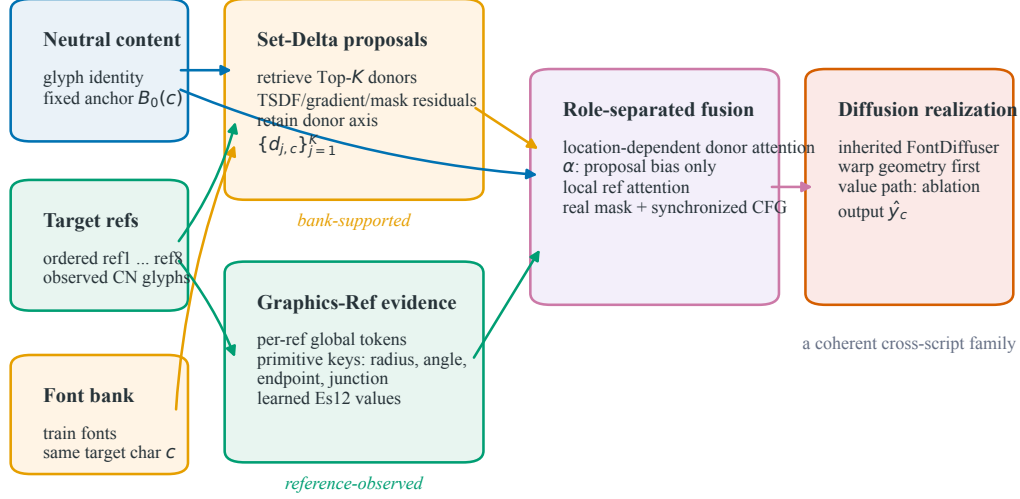


Figure 1: HR-Font separates two evidence roles. Retrieved anchor-relative residuals propose bank-supported structure; graphics-addressed reference features supply observed appearance. The denoiser fuses both while neutral content anchors identity.

- We introduce Set-Delta, which preserves normalized, anchor-relative same-character candidates until spatial aggregation, allowing each glyph region to select the most useful structural proposal.
- We introduce Graphics-Ref, which uses graphics primitives to find correspondence and learned reference features to deliver local appearance, retaining terminals, contrast, and brush texture.
- We deliver a multidimensional evaluation suite spanning identity, geometry, local style, family compatibility, and perceptual quality, with attributes organized by what the inputs reveal.

2 RELATED WORK

Cross-script font transfer. FTransGAN transfers a few observed glyphs across languages with multi-level attention (Li et al., 2021), and FCAGAN combines full-domain convolutional attention with a multi-layer perceptual discriminator (Zhao et al., 2024). DRA-font studies cross-lingual synthesis through diffusion-regularized adversarial learning and enhanced attention (Wu et al., 2026). Unlike these direct transfer systems, HR-Font casts cross-script generation as bank-supported completion guided by observed reference appearance.

Few-shot font generation. FontDiffuser casts one-shot font generation as conditional denoising and introduces multi-scale content aggregation and style contrastive refinement (Yang et al., 2023). DG-Font uses deformable feature transport to preserve structure under unsupervised font translation (Xie et al., 2021). FSFont learns spatial correspondence between content and fine-grained local styles through cross-attention (Tang et al., 2022), while VQ-Font combines similarity-guided global style with quantized local components (Pan et al., 2023). Local style and spatial alignment matter across these systems. HR-Font assigns correspondence and appearance to different representations: graphics descriptors align cross-script regions, while learned values carry the target family’s visual character.

Font-bank content priors. CF-Font projects content into a linear space of corresponding glyph features from basis fonts, with blending weights determined by font-level similarity (Wang et al.,

2023). This is the closest precedent for same-character bank fusion. HR-Font preserves individual donor residuals relative to one neutral anchor until local aggregation and separates structural proposals from target-reference appearance. We aggregate anchor-relative residual candidates rather than fused absolute glyph features. The result is a spatially adaptive variation prior that can use different donor evidence for different regions of the same glyph.

Geometric representation and decoupled conditioning. Signed distance representations support high-quality vector-font reconstruction and synthesis (Xia et al., 2023); HR-Font uses truncated signed distance fields as a normalized carrier for comparable glyph variation. Decoupled image conditioning has also appeared outside font generation (Ye et al., 2023). HR-Font applies role-specific conditioning to cross-script typography: proposal geometry pairs with observed appearance in one generation system.

3 PROBLEM FORMULATION

Let c denote a target character and x_c^0 its rendering in a fixed neutral font. For an unseen target font, the model observes an ordered reference set $R = \{r_i\}_{i=1}^n$, where $1 \leq n \leq 8$ during training and $n = 8$ in the main evaluation. The references and target may belong to different scripts. A training bank $\mathcal{B} = \{B_j\}_{j=1}^M$ contains fonts that render the target character and the reference character inventory. The target font is excluded from the bank.

We seek a generator

$$\hat{y}_c = G_\theta(x_t, x_c^0, \mathcal{D}_c(R), \mathcal{R}(R)), \quad (1)$$

where x_t is a noisy target image, $\mathcal{D}_c(R)$ is a set of candidate structural changes supported by the bank, and $\mathcal{R}(R)$ is appearance evidence extracted only from the observed target-font references. This formulation supports multiple compatible designs while keeping character identity anchored to x_c^0 .

4 METHOD

4.1 REFERENCE-CONDITIONED DONOR PROPOSALS

A frozen style encoder E_s maps each reference to a global descriptor h_i . Precomputed bank descriptors define a font-level similarity score $s_j(R)$. We retrieve the top- K donors and compute

$$\alpha_j = \frac{\exp(s_j(R)/\tau)}{\sum_{\ell \in \mathcal{J}(R)} \exp(s_\ell(R)/\tau)}, \quad j \in \mathcal{J}(R), \quad |\mathcal{J}(R)| = K. \quad (2)$$

α ranks the retrieved structural proposals and enters spatial aggregation as a logit prior. The main configuration uses $K = 10$ and $\tau = 0.07$; both values are frozen before formal evaluation.

4.2 SET-DELTA VARIATION PRIOR

Let $\phi(I)$ contain a clipped signed distance field, spatial gradients, and an ink mask computed from glyph image I . For every retrieved donor, we construct a same-character residual from the fixed neutral anchor:

$$d_{j,c} = \mathcal{N}(\phi(B_{j,c}) - \phi(x_c^0)), \quad (3)$$

where $\mathcal{N}$ performs scale and radius normalization and clips distances to eight pixels. The common anchor represents each candidate as a change from neutral content rather than as absolute donor geometry.

An encoder maps each residual to multi-scale candidate features $e_{j,p}^s$ at spatial position p and scale $s \in \{48, 24\}$. The current denoising and content state supplies queries q_p^s , while candidate geometry supplies keys $k_{j,p}^s$. We retain the donor axis and aggregate separately at each position:

$$a_{j,p}^s = \text{softmax}_j \left(\frac{\langle q_p^s, k_{j,p}^s \rangle}{\sqrt{d}} + \lambda_\alpha \log(\alpha_j + \epsilon) \right), \quad (4)$$

$$m_p^s = \sum_{j \in \mathcal{J}(R)} a_{j,p}^s e_{j,p}^s. \quad (5)$$

The main implementation maps m^s to a displacement field that warps content features through the existing deformable path. D7 evaluates residual-value injection as an alternative path.

The distinction from pre-averaged Delta is operational: a mean produces one global direction before seeing the target location, whereas Equation 5 permits different parts of one glyph to select different supported variations.

4.3 GRAPHICS-INFORMED LOCAL REFERENCE ATTENTION

Global style cues remain available as a sequence of per-reference pooled descriptors $\{h_i\}$ rather than their mean. To retain local evidence, we additionally extract a 12×12 frozen intermediate map $F_i = E_s^{12}(r_i)$ and pool it to 4×4 tokens per reference.

For each local region u , we compute a graphics descriptor

$$g_{i,u} = [\text{ink}, \text{radius}, \cos(2\vartheta), \sin(2\vartheta), \text{endpoint}, \text{junction}]_{i,u}. \quad (6)$$

The descriptor comes directly from each observed reference image and gives every local appearance value a reusable cross-script address. Keys and values have deliberately different roles:

$$k_{i,u} = W_k g_{i,u}, \quad v_{i,u} = W_v F_{i,u}. \quad (7)$$

Queries derived from the target content and denoising state attend over all valid (i, u) pairs. Graphics primitives propose cross-script correspondence; learned features carry the target font’s local appearance. A padding mask supports variable n and classifier-free guidance drops global references, local references, retrieval bias, and the Delta branch synchronously.

4.4 TRAINING OBJECTIVE AND IMPLEMENTATION BOUNDARY

We inherit the FontDiffuser denoising backbone and its epsilon-prediction objective:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{y, \epsilon, t} [\|\epsilon - \epsilon_\theta(x_t, t, x_c^0, \mathcal{D}_c(R), \mathcal{R}(R))\|_2^2]. \quad (8)$$

The total objective retains the existing perceptual reconstruction and deformation-offset regularizers, $\mathcal{L} = \mathcal{L}_{\text{diff}} + \lambda_{\text{perc}} \mathcal{L}_{\text{perc}} + \lambda_{\text{off}} \mathcal{L}_{\text{off}}$, with the same coefficients in every matched arm. The style and content encoders remain frozen and are consumed through versioned offline caches. R1–R3 add no local style loss. All new trainable projections use an initialization generator independent of the data and dropout random streams. The full method operates from the requested reference set and requires no additional target-font glyph collection.

5 EXPERIMENTS

5.1 QUESTIONS AND HYPOTHESES

The evaluation answers four preregistered questions:

1. Does replacing legacy feature Delta with canonicalized geometry Delta reduce donor-appearance leakage (D2 versus D1 plus leakage probes)?
2. Does preserving the donor axis improve compatible structure relative to geometry mean-Delta (D3 versus D2)?
3. Do local reference values improve observable stroke attributes beyond per-reference global tokens (R2 versus R1)?
4. Do graphics keys add useful cross-script correspondence beyond equal-capacity learned local attention (R3 versus R2, R4, and FSFont-style K/V)?

5.2 DATASET AND PROTOCOL

The current corpus contains 260 fonts after a preregistered ink-coverage exclusion, split by font into 228 training, 16 validation, and 16 test fonts. Protocol A renders 96×96 RGB glyphs using one font size per font, a neutral Noto Sans CJK Regular content face, 295 target characters, and a

Table 1: Completed current-system diagnostics on 752 glyphs. E12 scores capture family compatibility, while L1 and LPIPS measure pixel fidelity. Higher is better for ϕ SC-R and membership; lower is better for L1 and LPIPS.

Method	ϕ SC-R $\uparrow$	Membership $\uparrow$	L1 $\downarrow$	LPIPS $\downarrow$
F0 no RSI@100k	0.6719	0.5180	0.0707	0.1573
F1 official RSI@80k	0.6682	0.5314	0.0694	0.1498
F2 old mean-Delta@80k	0.7045	0.6010	0.0701	0.1452
F3 legacy Support@80k	0.7003	0.5948	0.0703	0.1437

338-character reference inventory. All primary comparisons use the same split, ordered reference set, effective batch size, data order, classifier-free guidance draws, and F0@100k parent checkpoint. The new method is trained with $n \sim \text{Uniform}\{1, \dots, 8\}$ and evaluated primarily with eight references. The exact data manifest hash, parent hash, optimizer, hardware, runtime, and total compute will be reported in Appendix A.

5.3 BASELINES

We prioritize faithful comparisons with FontDiffuser (Yang et al., 2023), the direct cross-script methods FTransGAN (Li et al., 2021) and FCAGAN (Zhao et al., 2024), and the bank-based CF-Font (Wang et al., 2023). FSFont (Tang et al., 2022) and VQ-Font (Pan et al., 2023) are reported only under documented protocols compatible with the cross-script setting. For every external baseline, a protocol card records its source and target scripts, reference budget, font-bank access, training corpus, and adaptation. We separately report bank-enabled matched controls under identical HR-Font resources, exposing the value of each information source under equal conditions. Internal controls include F0 (no RSI), F1 (official RSI source), F2 (old mean-Delta), D0–D6, and R0–R4.

5.4 METRICS

The primary endpoint is blind family/style preference, with confidence intervals clustered by font. Identity accuracy and artifact/readability judgments define non-inferiority gates. Geometry is measured with boundary Chamfer or SDF distance, hole and connected-component errors, endpoint error, skeleton-radius distributions, and slant/orientation distributions. Local blind labels cover terminals, ligatures, outline treatment, weight contrast, and dry-brush texture. Pixel metrics (LPIPS, SSIM, and L1) serve as diagnostics. The E12 score provides a targeted, lineage-defined measure of family compatibility alongside human preference and geometry metrics.

We stratify every attribute as (i) visible in the references, (ii) unobserved but represented in the training bank, or (iii) unsupported by both. This stratification measures observed-style transfer, bank-supported completion, and cases unsupported by either source.

5.5 MATCHED SCREENING AND FORMAL EVALUATION

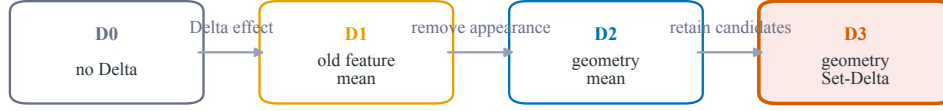
Figure 2 summarizes the staged design. Round D compares D0 no-Delta, D1 old feature mean, D2 geometry mean, and D3 geometry Set-Delta under the same R1 reference interface. After selecting D* using frozen validation rules, Round R compares R1 per-reference global tokens, R2 learned local K/V, and R3 graphics-keyed local values. R0, the legacy reference map, is included as a matched screening baseline. The 20k checkpoint provides an efficient interim decision point within the shared 40k schedule.

Formal test data are opened once after the configuration and validation thresholds are frozen.

5.6 MOTIVATING EVIDENCE FROM THE PREVIOUS IMPLEMENTATION

Table 1 reports completed diagnostics from the current system on 752 test glyphs. The old mean-Delta condition F2 improves the E12 compatibility proxies over F0 and F1, while pixel metrics disagree about the best method. The legacy support path F3 does not improve those proxies over F2. These observations motivate a geometry-centered Delta representation and a new local-reference interface.

Round D | hold R1 fixed; change only the variation prior



Round R | hold selected D* fixed; change only reference evidence

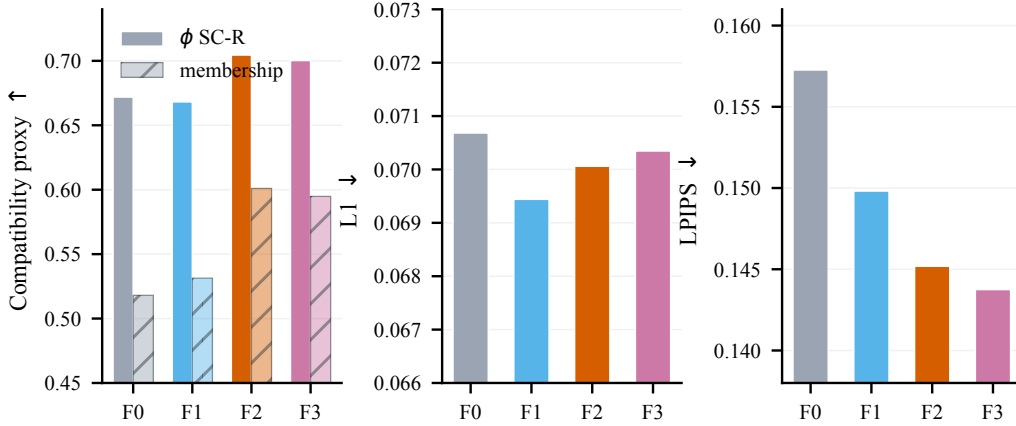


Focused controls that demonstrate each advantage

D3 vs D4: anchor-relative residual | D3 vs D6: target-character alignment
R3 vs R4 / FFont-style K/V: graphics semantics | D7: injection contribution

20k interim → paired continuation to 40k → formal evaluation

Figure 2: Claim-oriented experiment plan. Matched controls test residuals, character alignment, graphics semantics, and injection design while keeping the remaining system fixed.



Current-system diagnostics on 752 glyphs; metrics capture complementary properties.

Figure 3: Current-system evidence corresponding to Table 1. Delta delivers the strongest compatibility signal, while pixel metrics capture complementary behavior. This gap motivates HR-Font’s explicit separation of structural variation and observed appearance.

5.7 MAIN COMPARISON

[TBD: HR-Font improves blind family/style preference by X while preserving Y identity accuracy, and delivers the largest gains on Z geometry and local-style attributes.]

5.8 ABLATIONS AND CAUSAL DIAGNOSTICS

We additionally perform a wrong-donor × wrong-reference 2×2 intervention, clear the warp and value paths independently, exchange graphics keys from one font with learned values from another,

Table 2: Main cross-script generation results. We report means and 95% font-cluster bootstrap intervals; “–” marks results scheduled by the frozen evaluation protocol.

Method	Identity $\uparrow$	Style pref. $\uparrow$	Family compat. $\uparrow$	Chamfer $\downarrow$	Endpoint err. $\downarrow$	LPIPS $\downarrow$
FontDiffuser	–	–	–	–	–	–
CF-Font	–	–	–	–	–	–
FSFont	–	–	–	–	–	–
VQ-Font	–	–	–	–	–	–
F2 old mean-Delta	–	–	–	–	–	–
HR-Font	–	–	–	–	–	–

Table 3: Mechanism attribution. Matched comparisons test the geometry representation, candidate preservation, local appearance, and graphics correspondence.

Claim	Treatment	Direct control	Result
Canonicalized geometry package	D2 geometry mean	D1 old feature mean	–
Preserve candidate modes	D3 geometry set	D2 geometry mean	–
Anchor-relative residual	D3 residual set	D4 absolute set	–
Target-character use	D3 correct char.	D6 wrong char.	–
Local appearance evidence	R2 local K/V	R1 global refs	–
Graphics correspondence	R3 graphics keys	R2 learned keys	–
Key semantics	R3 correct keys	R4 shuffled keys	–
Dual injection	D7 warp+value	warp-only/value-only	–

and train probes to predict donor identity, weight, serif, and terminal classes from Delta alone. Together, these interventions reveal how each condition steers structure and appearance inside the generator.

6 SCOPE AND FUTURE OPPORTUNITIES

HR-Font targets compatible completion rather than reproduction of a single hidden design. Its structural vocabulary scales with the font bank and extends to rare, culturally specific, and emerging styles. Vector control points and learned stroke graphs offer future extensions for finer endpoint and topology control.

The current platform operates on 96×96 rendered glyphs and focuses on Chinese-to-Western and neighboring-script transfer. Extending the same evidence architecture to vector outlines, kerning, hinting, and text-line typography would support a broader cross-script type-design workflow. Our evaluation combines blind human judgments, structure-aware measurements, and representative challenging cases to map the strongest opportunities for that expansion.

7 CONCLUSION

We presented HR-Font, an evidence architecture for cross-script font completion: bank-supported structural choice plus reference-driven appearance personalization. Set-Delta retains same-character, anchor-relative geometry candidates until spatial aggregation, while Graphics-Ref separates primitive-based correspondence keys from learned local style values. Together, the two modules provide a scalable path from a few observed glyphs to a coherent cross-script family: **[TBD: summarize headline result and strongest mechanism finding]**.

AI USE STATEMENT

Generative AI tools played a significant role in organizing research ideas, reviewing claim boundaries, drafting text, and producing editable figure-generation code. They did not generate experimental measurements presented as evidence. All AI-assisted claims were checked against the versioned project documentation; cited literature was verified against primary paper or proceedings pages, and

	Content	Ref set	FTransGAN	FCAGAN	FontDiff.	CF-Font	HR-Font / Ours	GT
Ref-observable	RESULT PENDING	RESULT PENDING	RESULT PENDING	RESULT PENDING	RESULT PENDING	RESULT PENDING	RESULT PENDING	RESULT PENDING
Bank-supported	RESULT PENDING	RESULT PENDING	RESULT PENDING	RESULT PENDING	RESULT PENDING	RESULT PENDING	RESULT PENDING	RESULT PENDING
Failure case	RESULT PENDING	RESULT PENDING	RESULT PENDING	RESULT PENDING	RESULT PENDING	RESULT PENDING	RESULT PENDING	RESULT PENDING

Populate only with fixed content/ref/noise/sampler and a preregistered row-selection rule.

Figure 4: Editable qualitative-grid template. Each row fixes target, references, sampler, and noise; gray cells reserve matched outputs selected by the frozen protocol.

generated code and figures will be reviewed by the authors before submission. The authors take responsibility for the final text, claims, code, and artifacts.

ETHICS STATEMENT

Font generation can reduce repetitive design labor and improve script coverage, but it can also enable unauthorized imitation of proprietary typefaces. We will document the licenses and provenance of all fonts, exclude assets without suitable research permission, and avoid releasing target-font reconstructions where licensing forbids redistribution. Human evaluation, if conducted, will report participant instructions, consent procedures, compensation, and institutional requirements.

REPRODUCIBILITY STATEMENT

Appendix A specifies data splits, preprocessing, model variants, paired controls, randomness, metrics, and planned statistical analysis. The supplemental material will include versioned configuration files, run provenance, manifest hashes, evaluation scripts, and figure-generation scripts. Checkpoints and font assets will be released only where their licenses permit.

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A REPRODUCIBILITY DETAILS

Frozen configuration. Parent checkpoint: [TBD: SHA and artifact hash]. Code revision: [TBD: commit]. Data manifests: [TBD: train/val/test hashes]. Optimizer and scheduler: [TBD: exact values copied from resolved configuration]. Hardware and precision: [TBD: GPU model, count, precision, peak memory]. Runtime: [TBD: p50/p95 step time and total GPU hours].

Randomness and statistics. All training runs use the fixed random state 3407, paired data order, and paired conditioning draws. We report 95% confidence intervals by clustering resampling over fonts while preserving paired examples across methods.

Selection rule. Each arm must first satisfy frozen identity, artifact, and readability non-inferiority constraints. Among passing arms, the prespecified style/geometry endpoint determines continuation. If the difference is below the minimum relevant effect, the simpler arm wins. The numerical margins will be locked from validation-pilot variance before any 40k formal result is inspected.

B COMPLETE EXPERIMENT MATRIX

C HUMAN EVALUATION PROTOCOL

[TBD: Finalize sample size with a power analysis.] Raters will see the same target content and reference set for each paired comparison, with method order randomized and identifiers removed.

Table 4: Delta variants. All share the same parent, R1 interface, training data, draws, and endpoint.

ID	Definition	Identified comparison
D0	No Delta, shape-matched empty branch	Total method effect
D1	Old encoder-feature mean-Delta	Legacy Delta effect
D2	Geometry mean-Delta	D2–D1: canonicalized geometry package
D3	Geometry Set-Delta	D3–D2: candidate-axis value
D4	Same-bank absolute set	D3–D4: residual value
D5	Uniform retrieval weights	Learned retrieval bias
D6	Wrong-character residual set	Character alignment
D7	Warp-only/value-only/dual	Injection responsibility
D8	Global/local soft and hard assignment	Aggregation sensitivity
D9	Candidate concatenation + equal-capacity Conv	Attention capacity control

Table 5: Reference variants. All use the selected D* and the same observed reference images.

ID	Definition	Identified comparison
R0	Legacy mean spatial map	Historical reference path
R1	Per-reference pooled global tokens	Preserve reference axis
R2	Learned local Es12 keys and values	Local information increment
R3	Graphics keys, learned Es12 values	Graphics correspondence increment
R4	Matched shuffled primitive keys	Key-semantic intervention
R5	R3 + local style loss	Optimization fallback only

Separate questions will assess character identity, family compatibility, terminal/ligature/contrast attributes, artifacts, and overall preference. The protocol will include attention checks, ties, rater exclusion rules, compensation, and clustered uncertainty estimates.

D ADDITIONAL RESULT PLACEHOLDERS

- n-shot robustness for $n \in \{1, 2, 4, 8\}$: **[TBD: table/curve]**.
- Results by script and by ref-observability stratum: **[TBD: table]**.
- Donor leakage probe and key/value swap: **[TBD: table]**.
- Parameter count, FLOPs, memory, and runtime: **[TBD: table]**.
- Failure cases and bank-unsupported attributes: **[TBD: qualitative grid]**.